

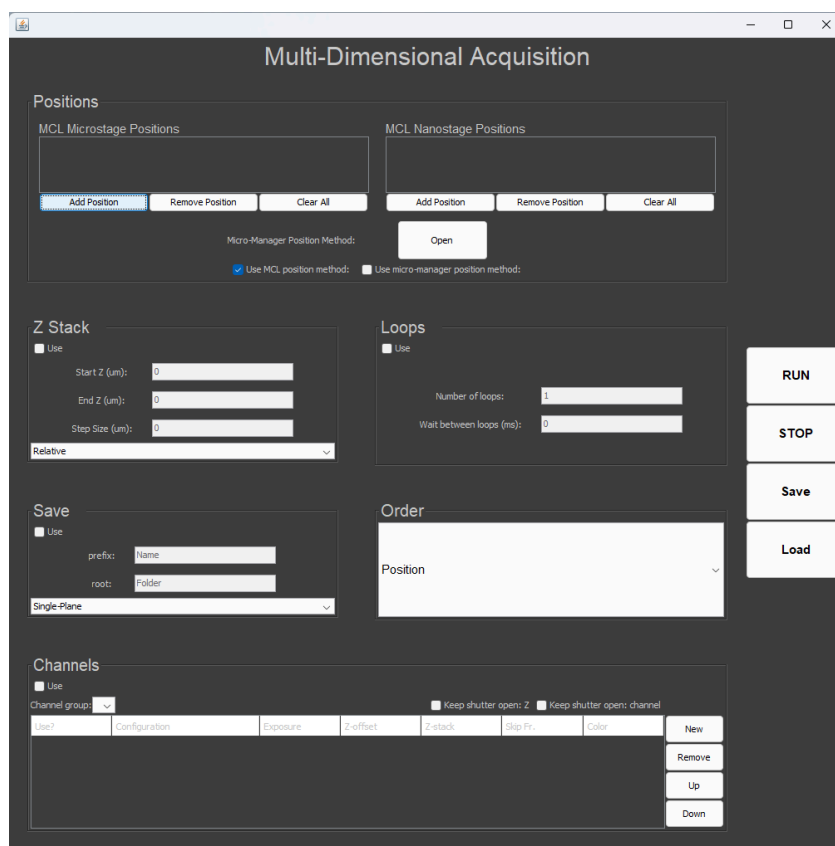
## Mad City Labs Multi-Dimensional Acquisition Plugin

Plugin and documentation created by Taylor Wang

### About This Document:

Micro-Manager is an open-source software that allows the control of microscope components. Micro-Manger has a feature called Multi-Dimensional Acquisition, which allows the user to collect images at selected positions using their microscope. However, the Mad City Labs RM21 Microscope has MicroStage movement and NanoStage movement. The Mutli-Dimensional Acquisition feature only allows for image acquisition using one of the stages, which is problematic for Mad City Labs customers. A solution was to create a custom plugin that allows for image acquisition with both stages.

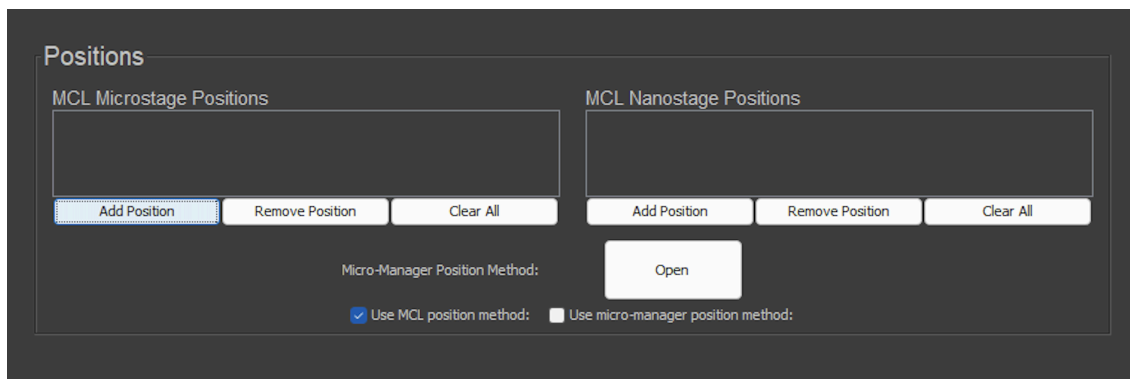
These instructions describe how to collect a Multi-Dimensional Acquisition using the Mad City Labs Multi-Dimensional Acquisition plugin (MCL Multi-Dim). All features from the original Micro-Manager Multi-Dimensional Acquisition are present in MCL Multi-Dim except auto-focus. MCL Multi-Dim offers a way to loop selected nano-positions within selected micro-positions, allowing for a more streamlined image acquisition experience.



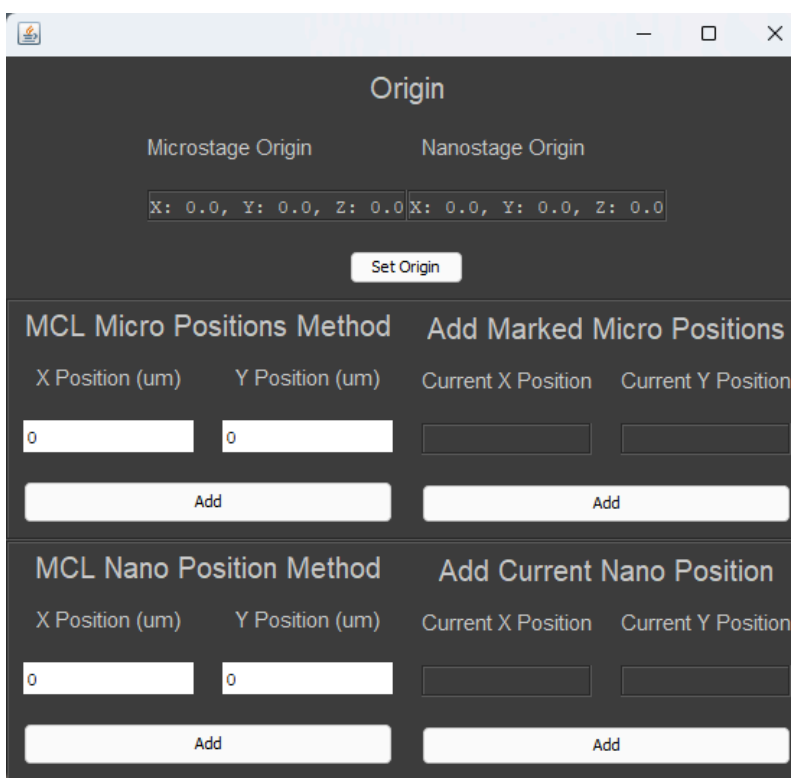
**Figure 1:** MCL Multi-Dimensional Acquisition Plugin Window

## MCL Position Method

The MCL Position Method allows the user to specify exact XY coordinates for the MicroStage and NanoStage to visit during the multi-dimensional acquisition.



**Figure 2:** *MCL Position Panel*



**Figure 3:** *MCL Position Window*



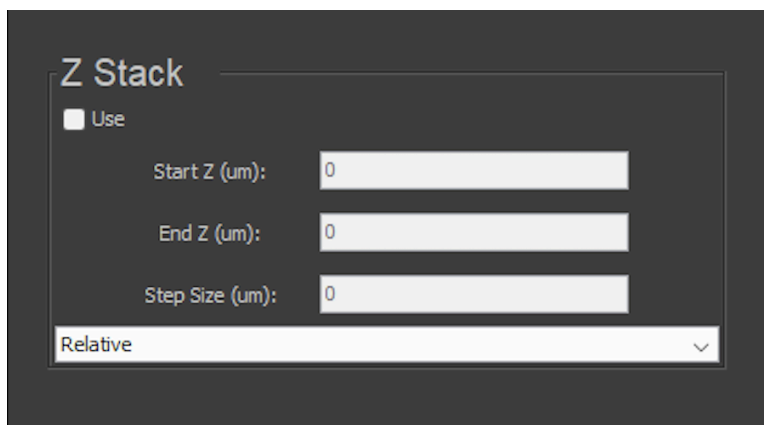
How to use MCL position method:

1. Open the MCL Multi-Dimensional Acquisition plugin located in the “Plugins” tab of the Micro-Manager main window.
2. To add a stage position, select the “Add Position” button beneath the MCL MicroStage Position panel or the MCL NanoStage Position panel. This will display a pop-up with three distinct areas: “Origin”, “MicroStage”, and “NanoStage”.
3. In the MicroStage or NanoStage area, a stage position can be added using a manual method or a marking method.
  - Manual Method: This method takes an X input and a Y input. Both these inputs refer to how far the stage will move from the current origin. For example, if “5” is inputted into the X position of the MicroStage, the stage will move 5 microns to the right. If “-5” is inputted into the X position, the stage will move 5 microns to the left. It is important to note the new position is NOT relative to the last added position. The new position is relative to an origin that does not change unless specifically told to by the “Set Origin” button.
  - Marked Method: This method displays the stage’s current X position and Y position, relative to the origin. Marking a position simply adds the current stage position to the position list. This means the stage control panel must be used to move the stage and mark different positions.
4. Once the X and Y inputs are set for the Manual Method or the stage is set for the Marked Method, press the “Add” button. This will add your desired position to the position list.
5. Set all other settings, like “Z-Stack” and “Time”.
6. Once all settings are set, click the “Run” button on the right side of the Multi-Dimensional Acquisition panel. The MicroStage will move to its first position and an image will be taken. Then the NanoStage will visit all the NanoStage positions and take an image at each position. After the NanoStage completes its loop, the MicroStage will move to its second position (if there is a second micro-stage position), and the NanoStage will perform the same loop. The process continues until all MicroStage positions are visited.

- ### Micro-Manager Position Method

## Z Stack

The Z stack panel allows the user to acquire images at different Z coordinates.



The screenshot shows the 'Z Stack' panel with a dark background. At the top left is the title 'Z Stack'. Below it is a checkbox labeled 'Use' which is currently unchecked. Further down are three input fields: 'Start Z (um):' with a value of '0', 'End Z (um):' with a value of '0', and 'Step Size (um):' with a value of '0'. At the bottom is a dropdown menu currently set to 'Relative'.

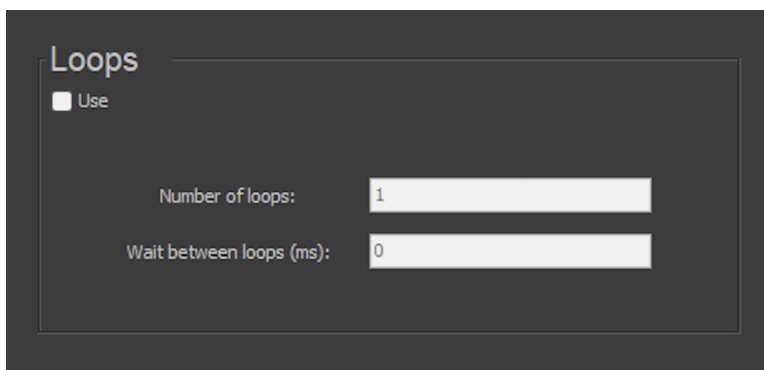
**Figure 5:** *Z Stack Panel*

How to use Z-stack:

1. Open the MCL Multi-Dimensional Acquisition plugin located in the “Plugins” tab of the Micro-Manager main window.
2. Click the “Use” check box within the Z Stack panel.
3. Enter a “Start Z” number, “End Z” number, and “Step Size” number. At each XY position, the stage starts at the Start Z position, then moves toward the End Z position in increments specified by Step Size. A picture is taken at each step/increment.

## Loops

The loops panel specify how many times the acquisition should repeat.



The screenshot shows the 'Loops' panel with a dark background. At the top left is the title 'Loops'. Below it is a checkbox labeled 'Use' which is currently unchecked. Further down are two input fields: 'Number of loops:' with a value of '1', and 'Wait between loops (ms):' with a value of '0'.

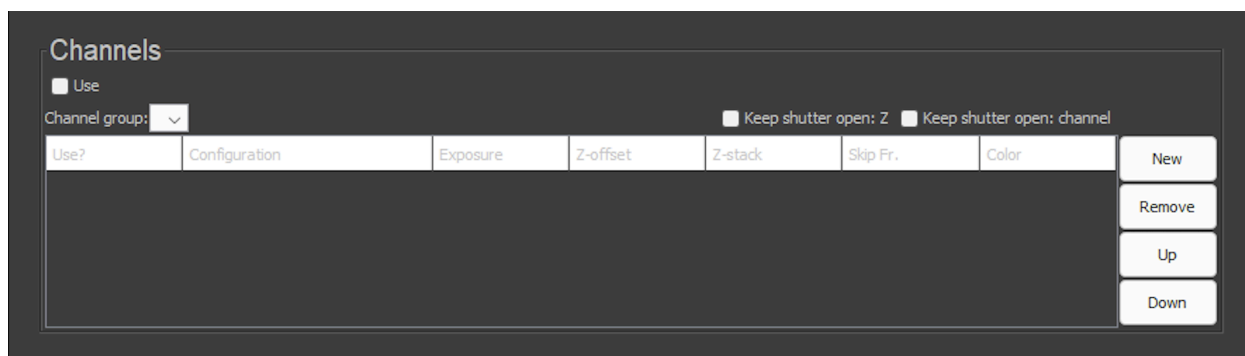
**Figure 6:** *Loops Panel*

How to use Loops:

1. Open the MCL Multi-Dimensional Acquisition plugin located in the “Plugins” tab of the Micro-Manager main window.
2. Click the “Use” check box within the loop panel
3. Enter a “Number of loops” number and a “Wait between loops” number. The number of loops specifies how many times the acquisition should repeat. The wait between loops specifies how long the acquisition should wait between each loop.

## Channels

Channels specifies the laser group used, the presets used, and the order in which the presets appear in the acquisition. The laser group is the list of all lasers that can be used in a preset. A preset is a list of which lasers are on and off. Multiple presets can be created to acquire images of the same position but with different lasers on.



**Figure 7:** *Channels Panel*

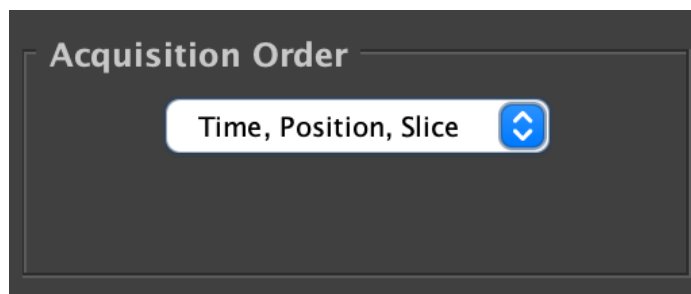
How to use channels:

1. First, a channel group must be made. Go to the main Micro-Manager window and focus on the huge table with the sections labeled “Group” and “Preset”. Find the “+” sign next to the word “Group:” at the bottom of this table. Click on the “+” to bring up the Group Editor.
2. Add all desired lasers to the group by selecting the checkbox for the emission property of each laser.
3. Name the group and press “OK”.
4. Now channel presets must be made. Find the “+” sign next to the word “Preset:” at the bottom of the huge table. Click on the “+” to bring up the Preset Editor.
5. Select which lasers you want on, and which lasers you want off.

6. Name the Preset and press “OK”.
7. Repeat the preset process to create different combinations of lasers.
8. Once the group and all presets are created, open the MCL Multi-Dimensional Acquisition plugin located in the “Plugins” tab of the Micro-Manager main window
9. Click the “Use” checkbox within the channels panel.
10. Click the “Channel Group” dropdown menu to select the group that was just created.
11. Once the group is selected, click the “New” button on the right side of the channels panel.  
This will populate the table with the first preset you created. Click the “New” button until all presets appear on the table.
12. Click the “Use” checkbox next to each laser preset you want to use in the acquisition.
13. Set the order of the presets by putting the first preset at the top and the last preset at the bottom using the “Up” and “Down” buttons in the channels panel. At each position during the acquisition, the lasers corresponding with the first preset will turn on and an image will be acquired. Then the lasers corresponding to the second preset will turn on and an image will be acquired. The same process will repeat until all presets are visited.

## Order

The acquisition can execute actions in different orders. The possible actions are position, Z stack, loops, and channel. For example, an order can be Loops > Position > Z Stack > Channel. This means the acquisition will visit each channel before moving on to the next Z Stack position. And it will visit each Z Stack position before moving on to the next XY position. And finally, it will visit each XY position before looping again.



**Figure 8** *Acquisition Order Panel*

How to use order:

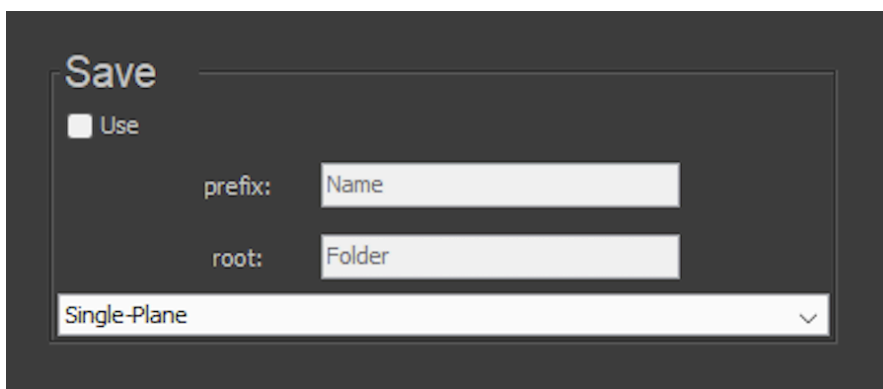
1. Open the MCL Multi-Dimensional Acquisition plugin located in the “Plugins” tab of the Micro-Manager main window.
2. Make sure to click the “Use” check box for each setting you want to use in the acquisition. Once the “Use” check box is selected for a setting, the Order drop-down

menu will populate with the corresponding action. For example, clicking “Use” for Z Stack will cause the order drop-down menu to populate with “slice”.

3. Click on the “Order” drop-down menu and select the order you want

## Save Acquisition

The save panel allows the user to save their acquisition with a specific name and within a specific folder.



**Figure 9:** *Save Acquisition Panel*

How to save acquisitions:

1. Open the MCL Multi-Dimensional Acquisition plugin located in the “Plugins” tab of the Micro-Manager main window.
2. Click the “Use” check box within the Save panel.
3. Enter a prefix, which will be the name of the acquisition. Then enter the root, which is the path where the acquisition file will be saved.
4. The acquisition images will be saved in the specified folder after the acquisition is completed.

## Save and Load Acquisition Settings

Save and load allows the user to save acquisition settings and position settings.

How to save settings:

1. Once all acquisition settings are set, click the “Save” button. The “Save the acquisition settings file” window will appear.
2. Enter a name for the acquisition settings and select an appropriate folder. Click save.



3. Next the “save the position settings file” window will appear.
4. Enter a name for the position settings and select an appropriate folder. Click save
5. The settings should be saved in the folder you specified

#### How to load settings:

1. Click on the “Load: ” button. The “Load acquisition settings” window will appear. Find the desired acquisition settings file and press open. The Multi-Dimensional Acquisition UI should populate with the settings in the file.
  - Note: You may need to switch the File Type from Acquisition settings to All Files to actually find your file
2. Next, the “Load position settings” window will appear. Find the desired position settings file and press open. The Multi-Dimensional Acquisition position list should populate with the positions saved in the file.